

Sai Maanasa

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PROFESSIONAL SUMMARY

- **Data Engineer with 4+ years of experience** building scalable data platforms using **Python, PySpark, SQL, and Snowflake** across AWS and Azure cloud environments.
- **Expert in end-to-end data engineering** including ETL/ELT pipelines, data warehousing, schema design, and performance optimization for high-volume analytical workloads.
- **Strong cloud-native engineering experience**, leveraging **AWS (S3, Glue, Lambda, Redshift, EMR)** and **Azure (ADF, ADLS, Synapse)** to deliver reliable, cost-efficient data solutions.
- **Hands-on experience with Generative AI-driven data workflows**, developing backend services and automation using **FastAPI, REST APIs, Azure OpenAI, and LLM frameworks**.
- **Proven strength in data quality, observability, and automation**, implementing validation frameworks, monitoring, and logging with **Splunk, Grafana, and CloudWatch**.
- **Experienced in data modernization, CI/CD, and orchestration**, using **Airflow, AWS Step Functions, Control-M, and CodePipeline** to enable scalable batch and streaming pipelines.
- **Delivered large-scale, domain-driven data solutions** across **sustainability (Scope 3 emissions), healthcare (HIPAA-compliant claims), and financial fraud analytics**, leveraging **Big Data, cloud lakehouse architectures, real-time streaming, and ML pipelines** to enable regulatory compliance, real-time insights, and data-driven decision-making.

SKILLS

- **Cloud Platform:** AWS (S3, Glue, Lambda, EMR, RDS, Redshift), Athena, CloudWatch, Azure, GCP, Snowflake, Teradata
- **Programming:** Python, SQL, PL/SQL, PySpark, DBT, Neo4j, R, Java, Powershell, Linux
- **Big Data & Migration Tools:** Hadoop, Hive, Spark SQL, AWS Glue Studio, Snowpipe, Data Frame API
- **Database Management & Tools:** PL/SQL, Oracle 19c, PostgreSQL, MongoDB, MySQL, DynamoDB, CosmosDB
- **Visualization and Analytical Tools:** Tableau, Power BI, Microsoft Office (Advance Excel, PowerPoint and Word)
- **Statistical Modeling:** A/B testing, Regression Analysis (Linear & Logistic), Classification (Decision Trees, SVM, k-NN, Naïve Bayes, Neural Networks, Gradient Boosting, Random Forest)
- **Agile & Collaboration:** Scrum, Sprint Planning, Backlog Grooming, Code Reviews, Mentorship
- **Security & Compliance:** Microsoft Cloud Background Check, SOX, HIPAA, PHI Masking, RBAC, Audit Logging, Encryption (KMS, TLS), Data Retention, IAM Policies

PROFESSIONAL EXPERIENCE

XNODE INC

Data Engineer

Jun 2025 – Till Date

Dallas, TX

- Initiated the **Data Modernization** project by leveraging **Python, PySpark, and Snowflake** to support advanced **data warehousing solutions**.
- Engineered a **scalable ETL pipeline** using **Python** and **SQL** to ingest, transform, and load **Azure DevOps** work-item data, enabling **low-latency analytics** and improving **data availability** for **downstream applications**.
- Built an **automated QA data-validation agent** that crawls **source URLs** and generates **pytest cases** to identify **data-quality issues**, increasing **test coverage** and attracting interest from **two clients**.
- Developed **backend data services** using **FastAPI** and **REST APIs** to support **LLM-driven workflows**; implemented **data-access optimizations** that increased **user engagement** by **30%**.
- Configured **Splunk** and **Grafana** dashboards to enhance **data observability**, improve **logging accuracy**, and reduce **incident detection and response times**.
- Created a **Developer Agent platform** using **Azure OpenAI, LangGraph, and Claude** to automate **code evaluation, issue diagnosis, and repository integration**, improving **engineering throughput** by **50%** and accelerating **issue resolution** by **40%**.
- Drove adoption of **data automation solutions** by developing **specialized agents** for **Jira-Postgres sync, AML screening pipelines, Instagram workflow automation, and Databricks-based risk analysis**, boosting **process efficiency** and **client satisfaction**.
- Contributed more than **15,000 lines of high-quality Python code** that improved **system scalability, automated data processes**, and supported **new client onboarding**—earning **recognition from the CEO and Technical PM**.

WIPRO

Data Engineer

Jan 2020 – Aug 2023

India

- Established robust **data warehousing solutions** by creating **Azure Data Factory pipelines** for the movement of files to **Azure Data Lake Storage**.
- Enhanced **operational efficiency** by reducing **turnaround time** through the development of **data pipelines** that moved data between **Azure Synapse tables and Snowflake**, reinforcing **ETL processes**.

- Developed **parameterized pipelines** with **reusable libraries** to support **scalable data architecture** development for **diverse workflows and data sources**.
- Migrated **SSIS packages** to **Azure Data Factory Pipelines** and deployed them in **production**, exemplifying effective **data architecture construction**.
- Created a **Tableau dashboard** to track **ML model accuracy and performance**, supporting **data quality assurance** with **weekly retraining insights**.
- Tested **data architectures** by orchestrating **weekly table refreshes** on **Hadoop** using **Hive, Impala, and Oozie**, ensuring **system reliability**.
- Maintained **data architectures** by executing **Presto and SQL queries** on **large datasets** and building **interactive Tableau dashboards** to track **key performance indicators (KPIs)**.
- Spearheaded the migration of **legacy Ab Initio ETL pipelines** to **Snowflake**, utilizing **AWS Glue, S3, and Kafka**, significantly improving **performance, maintainability, and scalability**.
- Designed **optimized Snowflake schemas**, leveraging **clustering keys** and **partitioning strategies** for **high-volume data (inpatient/outpatient claims, transactions)**.
- Developed **SQL procedures, validation scripts, and materialized views** in **Snowflake and Oracle** to support **operational dashboards and compliance reporting**.
- Automated **data workflows** using **AWS Step Functions, Control-M, and Airflow**, enabling **fault-tolerant, scheduled batch and stream processing**.
- Implemented **CI/CD pipelines** using **AWS CodePipeline, Python, and Shell scripts** to streamline **deployment and environment setup**.
- Collaborated with **data governance and compliance teams** to ensure **HIPAA-compliant data handling** and **secure architecture** across **ingestion and reporting layers**.

KL University

Jan 2019 - Jan 2020

Teaching Assistant

India

- Provided **technical guidance and troubleshooting** for **70 students**, effectively aiding the instructor in delivering a comprehensive **Python course**.
- Built and maintained **automated grading and evaluation scripts** in **Python**, leveraging **unit tests, data validators, and logging** to assess **code correctness, performance, and edge-case handling at scale**.
- Developed and demonstrated 200 students **end-to-end Big Data workflows**, including **data ingestion** from **flat files and APIs**, **schema management** with **Hive**, **distributed querying** using **Spark SQL**, and **performance tuning** through **partitioning and caching**.

RELEVANT PROJECTS

Sustainability-Focused Emission Analysis (Big Data, Cloud, AI)

- Evaluated Scope 3 emissions from a vehicle portfolio using a bottom-up approach, analyzing Well-to-Tank and Tank-to-Wheel emissions to establish a rigorous baseline.
- Collaborated with the bank to develop a strategic plan aimed at reducing emissions by 20% over the next decade, providing actionable pathways for working with automotive companies to achieve net-zero goals.

Healthcare Claims Data Automation (AWS Glue, Airflow, Redshift, Python, HIPAA)

- Engineered secure ETL pipelines using AWS Glue, Airflow, and Python to consolidate healthcare claims from multiple upstreams into a compliant analytics layer.
- Enabled HIPAA compliance via data encryption, audit logging, and de-identification policies to support regulatory data delivery.

Real-Time Compliance & Fraud Analytics (Azure, Kafka, Databricks, Power BI, MLflow, Delta Lake)

- Built a real-time fraud analytics platform leveraging Kafka, Delta Lake, and Azure Synapse to monitor financial transactions under 60 seconds.
- Integrated production-ready ML models using MLflow for anomaly detection and embedded model outputs into Power BI dashboards with full RBAC controls.

Cloud Data Lakehouse for Unified Analytics (Azure Synapse, Databricks, Spark, SCD, Parquet)

- Developed a scalable data lakehouse for procurement, finance, and inventory systems using Azure Synapse, Spark, and Parquet.
- Implemented SCD Type 2, schema validation, and gold/silver/bronze zone partitioning to optimize performance and audit lineage.

EDUCATION

The University of Texas at Arlington

Master of Science, Computer Science

K L University, India

Bachelor of Technology, Computer Science and Engineering

CERTIFICATIONS

- AWS Certified Data Engineer Associate
- AWS Certified Cloud Practitioner
- Pega Certified (Senior System Architect (PCSSA) v8.6, System Architect (PCSA) v8.4, Robotics System Architect (PCRSA))
- IBM Data Science Specialization | Statistics with Python (University of Michigan) | Mathematics for Data Science
- CISCO NETACAD – Operating Systems Bridge Course